

React JS Practical & Conceptual Examination

Total Marks: 30

Duration: 90 Minutes

Instructions

- Answer all questions in a Microsoft Word document.
- Write code wherever required.
- Explain your approach clearly.
- Do not copy from the internet.
- Mention assumptions if required.

Section A – React Fundamentals (10 Marks)

Q1. (2 Marks)

Explain the complete rendering process when a user clicks a button that updates a counter using `useState`.

Q2. (2 Marks)

What will be the output of the following code? Explain why.

```
function App() {
  const [count, setCount] = useState(0);

  return (
    <>
      <h2>{count}</h2>
      <button onClick={() => {
        setCount(count + 1);
        setCount(count + 1);
      }}> Increment
    </button>
    </>
  );
}
```

- What value will be displayed?
- Why?
- How can you make it increase by 2?

Q3. (3 Marks)

Differentiate between Props vs State, `useEffect` vs `useState`, Controlled Component vs Uncontrolled Component. Give one real-time example for each.

Q4. (3 Marks)

A list is rendered without using the `key` property.

```
{students.map(student => (
  <StudentCard name={student.name}/>
))}
```

- Why is this wrong?
- What problems can occur?
- What should be used as the key?
- Why should array index be avoided?

Section B – Code Analysis (10 Marks)

Q5. (4 Marks)

Find and correct all mistakes.

```
function App() {
  const users = [];

  useEffect(() => {
    fetch("https://jsonplaceholder.typicode.com/users")
      .then(res => res.json())
      .then(data => {
        users = data;
      })
  })

  return (
    <div>
      {
        users.map(user => (
          <p>{user.name}</p>
        ))
      }
    </div>
  )
}
```

- Explain every mistake.

Q6. (3 Marks)

Predict the output.

```
function Parent() {
  const [name, setName] = useState("John");

  return (
    <>
      <Child />
      <button onClick={() => setName("David")}>
        Change
      </button>
    </>
  )
}

function Child() {
  console.log("Child Rendered");
  return <h1>Hello</h1>
}
```

- Will Child render again?
- Why?
- How can you prevent unnecessary rendering?

Q7. (3 Marks)

Write the folder structure for a medium-sized React application including Login, Dashboard, Products, Cart, API Calls, Reusable Components, Custom Hooks, Context API, and Utilities. Explain why this structure is better than keeping everything inside the src folder.

Section C – Scenario Based Questions (10 Marks)

Q8. (4 Marks)

You are developing an E-commerce application with Login, Dashboard, Product Listing, Product Details, Cart and Logout modules.

- Which pages should be protected?
- Where will you store the logged-in user information?
- Which React concepts will you use to build this application?
- Draw the component hierarchy.

Q9. (3 Marks)

A Weather API is taking 8 seconds to respond. How will you improve the user experience?

Q10. (3 Marks)

You have five components using the same API call. Without writing code, explain how you would reuse the API logic in React.

Evaluation Criteria

React Fundamentals – 10 Marks

Code Analysis & Debugging – 10 Marks

Scenario-Based Thinking – 10 Marks

Total – 30 Marks